3.6.1 Application of Cubic Identities

I. Using the identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Example 3.13

Expand $(x+4)^3$

Solution:

Comparing $(x+4)^3$ with $(a+b)^3$, we get $a = x, b = 4$

We know $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$$(x+4)^3 = (x)^3 + 3(x)^2(4) + 3(x)(4)^2 + (4)^3 \quad \text{(replacing } a, b \text{ values)}$$

$$= (x)^3 + 3x^2(4) + 3(x)(16) + 64$$

$$(x+4)^3 = x^3 + 12x^2 + 48x + 64$$

$(4)^2 = 4 \times 4 = 16$

$(4)^3 = 4 \times 4 \times 4 = 64$

Try to expand this by using

$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

Example 3.14

Find the value of $(103)^3$

Solution:

Now, $(103)^3 = (100+3)^3$

Comparing this with $(a+b)^3$, we get $a = 100, b = 3$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 \quad \text{replacing } a, b \text{ values,}$$

$$(100+3)^3 = (100)^3 + 3(100)^2(3) + 3(100)(3)^2 + (3)^3$$

$$= 1000000 + 3(10000)(3) + 3(100)(9) + 27$$

$$= 1000000 + 90000 + 2700 + 27$$

$$(103)^3 = 1092727$$

II. Using the identity $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

Example 3.15

Expand: $(y-5)^3$

Solution:

Comparing $(y-5)^3$ with $(a-b)^3$, we get $a = y, b = 5$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$(y-5)^3 = (y)^3 - 3(y)^2(5) + 3(y)(5)^2 - (5)^3$$

$$= (y)^3 - 3y^2(5) + 3(y)(25) - 125$$

$$(y-5)^3 = y^3 - 15y^2 + 75y - 125$$

Try to expand this by using

$(a-b)^3 = a^3 - b^3 - 3ab(a-b)$

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